Riley Beenders

PRODUCT DESIGN | DESIGN SYSTEMS | END-TO-END SHIPPED INTERFACES

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PROFESSIONAL SUMMARY

Engineer and designer who takes ambiguous problems through to shipped experiences that people use every day, from customer-facing products to plant-floor software and a portfolio site with its own design system. Designs around the people using the work and their constraints, partners closely with engineering and leadership, and explains the reasoning behind each decision.

CORE EXPERTISE

Product & UX Design: End-to-end ownership from problem framing to launch; user flows; interfaces designed per audience (operators, packing technicians, management); touch-screen kiosk interfaces; requirements for remote-operation software; plug-and-play customer setup workflows; industrial design; user guides and technical writing

Design Systems & Front End: Design tokens and theming; type, spacing, and color scales; motion systems with reduced-motion support; focus-visible states and ARIA labels; React, Next.js, TypeScript, framer-motion; HTML and CSS; responsive layouts down to phone width

AI-Assisted Workflow: Claude Code as a core build tool; repository agent skills for resume tailoring, documentation sync, and motion decisions; design critique and anti-pattern detection against written product and design context; Playwright screenshot capture and visual verification

Collaboration & Delivery: Cross-functional work with engineering, production, and leadership; external software partners; worldwide project collaboration; reviews of status, constraints, and budgets with global leadership; Kanban in Microsoft Teams; Git and GitHub; hiring manager and technical interviews

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below continued as Lead R&D Engineer for the same product line, now Proteor Print.

- Designed and built ExtrusionStudio, the plant's production system, around the people who use it: a projector view for the floor, a touch screen per line, a packing-bench panel, and a statistics TV for management. Live on all three filament extrusion lines since October 2026, rebuilt from a LabVIEW proof of concept.
- Replaced full-sheet label printing with print-on-demand thermal labels at the packing bench, removing up to about 30% of label waste per run.
- Fixed a lagging, disconnecting packing-bench touch screen during rollout by testing one layer at a time: halving the animation frame rate, checking for thermal throttling, and swapping in a fresh Raspberry Pi changed nothing; a direct Ethernet run fixed the dropouts and the label delay, and running the 1440p screen at 1080p removed the lag.
- Led the design and development of the ICARUS-Lite, a compact industrial printer that automates prosthetic and orthotic production with minimal customer setup, owning its mechanical, electrical, and software design; gave it a bolt-less shell with seamless chamfered geometry that carries the POSEIDON's design language and a plug-and-play customer workflow.
- Owned the ICARUS-Lite through commercialization against unit goals of 60 in 2025 and 100 in 2026.
- Contributed the machine requirements and integration for ProteorPrintOS, built with an external software partner on SimplyPrint, giving users remote operation, live video, AI-assisted failure alerts, and material, waste, and outcome metrics.
- Co-developed a custom hotend with an outside partner, improving reliability and reducing cost for our materials.
- Developing an offline version of the ICARUS-Lite and ICARUS-Pro control screen with over-the-air automated updates, so customers can run without a Wi-Fi or Ethernet connection.
- Serve as the go-to contact for customer troubleshooting beyond what support can resolve, across mechanical faults, electrical faults, and most often user error, covering how to mitigate the error, move forward, and learn from it.
- Took the Coplay location's Microsoft Dynamics 365 Business Central ERP from evaluation to live production in five weeks, migrating more than 2,000 items and 150+ products and syncing SKUs with Proteor's global locations; the whole facility works in it every day.
- Present project status, constraints, and budgets at Proteor's monthly USA review with leadership from its global locations, and write incident reports for leadership.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.

- Managed and expanded the division's IT infrastructure after the acquisition, standardizing software and internal processes to align with Proteor's company-wide systems.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Across these roles, wrote customer proposals, scopes of work, and estimates, and began tracking project budgets.

Chief Technology Officer — January 2023 - January 2025

- Oversaw the product-development pipeline and technology strategy while keeping hands-on responsibility for mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Led the design and launch of Gen3 POSEIDON, the flagship's production redesign, combining a 1 m^3 build volume, filament-and-pellet dual extrusion, and braked Z drives that prevent uncontrolled motion during power loss.
- Designed and hosted a self-developed ERP/CRM on a redundant Kubernetes cluster that brought orders, inventory, tasks, projects, and customer interactions into one tool, generating BOMs dynamically and linking design data directly to purchasing.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.
- Removed a manufacturing bottleneck with independently testable electronics enclosures that installed in minutes, coordinating custom harnesses, cut panels, and hardware improvements with vendors.

Industrial Engineer — April 2021 - June 2022

- Delivered THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, carrying it from CAD, BOM, and sourcing through fabrication, assembly, and firmware tuning to customer support and operator training.
- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.

Engineering Technician — May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and improvement opportunities.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, and coordinated final device distribution with school administrators.

Blue Mountain Ski Resort - Ski Lift Operator (seasonal) | November 2019 - January 2021 | Operated lifts and assisted thousands of guests with ticketing, weather, and slope information; performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Carver House Homeowners Association - Treasurer | April 2026 - Present | Manage the banking and monthly insurance payments for an association of about 80 units, and help establish a community-focused member board in place of a property management company, remaining Treasurer through a mid-year restructure.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering — December 2024

Machine Learning Onramp, MathWorks	2026
MATLAB Onramp, MathWorks	2026
Simulink Onramp, MathWorks	2026
Simscape Onramp, MathWorks	2026
Lean Six Sigma Yellow Belt, Educate 360	2025
Lean Six Sigma White Belt, Educate 360	2025
Certificate in Engineering Design and Tools, Penn State University	2023
Mechanical Engineering Certificate, Lafayette College	2019

SELECTED PROJECTS

RileyBeenders.com | Portfolio site and design system, rileybeenders.com/about-this-site — June 2026 - Present

Next.js, React, TypeScript, framer-motion, Vercel, Claude Code, Playwright

Problem: A PDF resume flattens every project into two lines. The site had to read like that resume at rest for a recruiter's ten-second scan, open into the evidence behind each line for a technical reader, and never turn into a marketing page.

- Designed the Blueprint Press system: paper, ink, hairline rules, a fixed blueprint grid, and one easing curve shared by every transition, with ten palettes and a dark mode built as a token swap on paper and ink rather than a second design.
- Linked every project from a line of the resume with its case study a click away, so the site reads at two depths: the surface for every visitor, the case studies for the technical reader.
- Prototyped the redesign as a preview route beside the live design, then promoted it and deleted the old one; motion stays subtle enough for a quick scan, with no scroll-jacking, and stops under reduced motion.
- Split content from design: one resume JSON became focused data files merged at build time, edited from a local schema-driven Studio, so content, palette, and typography change from a browser form without touching code.
- Wrote the product and design context (who visits, what a successful visit is, the constraints to preserve) as files that AI agents read before any design work, with a critique and anti-pattern skill checking changes against them.

Interactive HTML Resume | Web layer for tailored resumes, rileybeenders.com — November 2026 - Present

HTML, CSS, JavaScript, Playwright

Problem: A tailored PDF has to stay ATS-safe and fixed in length, while a reader on screen may want more depth than the printed page holds.

- Wraps the same verified source in the site's styling: a sticky bar with section links that mark the one in view, a theme toggle, skill pills that dim every line not mentioning a clicked term, expandable bullet details, and a Print button that prints the exact page count of the PDF.
- Verified automatically in light, dark, and at 390px phone width, with JavaScript off, and page by page against the delivered PDF; hover motion runs only on devices that can hover, so a tap never leaves it stuck.

ExtrusionStudio | Plant measurement and production system, Proteor — September 2025 - Present

Python, Starlette, Server-Sent Events, SQLite, Raspberry Pi kiosks

Problem: The only record of a spool's diameter was whatever an operator wrote down, and the plant needed every spool judged against its tolerance with the result visible to the operator, the packing bench, and management at once.

- Runs each audience's panel as its own process, with only the admin settings allowed to talk to the gauges.
- Designed the packing-bench workflow: technicians approve, inspect, or reject each spool with the reasons recorded on its serial, then print its spool, box, and master case labels; each approved spool gets a QR label and an HTML certificate.
- Ties every reading and alarm to the spool it is actually wound on, not the one under the laser, and sends min, max, and last values per channel in each frame so a spike between frames still shows on the live chart.

BulkSSH | Windows desktop app, github.com/RileyBeenders/BulkSSH — October 2026 - Present

TypeScript, Electron, Node.js

Problem: Testing ExtrusionStudio's kiosks meant repeating the same commands on every device, one window at a time.

- Opens every device in a paged grid of live terminals with a full-size focus view, mirrors keystrokes to every ticked device, and asks for confirmation before a destructive command reaches several devices.
- Installs for one user with no admin rights or runs as a single portable file; saved passwords are optional and encrypted with the Windows account, and dropped connections reconnect by themselves.

RB's Auto App Updater | Windows utility — September 2026

PowerShell, WinGet, Windows Task Scheduler

Problem: Keeping dozens of Windows applications current meant visiting websites and running each installer by hand.

- Updates every application with one double-click and no UAC prompt after a one-time setup, and translates installer exit codes into plain-language statuses such as security blocked, restart needed, and application in use.
- Upgrades each package on its own so a single failure never stops the rest, writes detailed logs that make remote troubleshooting possible, and updates itself only after the new script's SHA-256 hash matches its version manifest.